

Platinum Single Atoms Supported on Nanoarray-Structured Nitrogen-Doped Graphite Foil with Enhanced Catalytic Performance for Hydrogen Evolution Reaction

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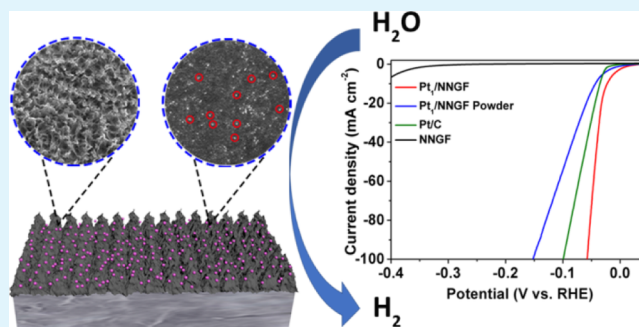
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Supporting Information

ABSTRACT: Platinum-based single-atom catalysts (SACs) are among the most promising candidates for the practical applications of electrochemical hydrogen evolution reaction (HER), but their catalytic efficiency remains to be further enhanced. Herein, a well-designed nanoarray-structured nitrogen-doped graphite foil (NNGF) substrate is introduced to support Pt SACs in Pt–N₄ construction (Pt₁/NNGF) for HER. Within NNGF, the constructed nanoarray-structured surficial layer for supporting Pt SACs could enhance the exposure of active sites to the electrolyte and improve the reaction and diffusion kinetics; meanwhile, the retained graphite structures in bulk NNGF provide not only the required electrical conductivity but also the mechanical stability and flexibility. Because of such double-layer structures of NNGF, stable Pt–N₄ construction, and binder-free advantages, the Pt₁/NNGF electrode exhibits a low overpotential of 0.023 V at 10 mA cm^{−2} and a small Tafel slope of 29.1 mV dec^{−1} as well as an excellent long-term durability.

KEYWORDS: nitrogen-doped graphite foil, plasma treatment, nanoarray structure, photochemical reduction, Pt single atoms, hydrogen evolution reaction



INTRODUCTION

With the increasing energy demand and worldwide environmental concerns, the exploration of reliable and renewable sources of clean energy becomes more and more urgent.^{1,2} The generation of hydrogen (H₂) fuel through water electrolysis from renewable electricity is believed to be the most promising pathway toward the sustainable economy.^{3,4} To realize the practical application of electrochemical hydrogen evolution reaction (HER), an economical, efficient, and durable catalyst is required to accelerate the catalytic kinetics and reduce the required overpotential.^{5,6} Among the developed electrocatalysts, Pt-based catalysts are the most active ones for HER catalysis because of the optimal binding energy of the Pt–H bond.^{7,8} Considering the rocketing price and scarcity of Pt, it is highly desirable to maintain or enhance its HER efficiency with as few dosages as possible. Currently, several strategies have been proposed to reduce the Pt loading amount in catalysts, including downsizing Pt nanoparticles, fabrication of Pt-based alloy and combination of Pt with nonprecious metal-based catalysts, and so forth.^{9–11} Among them, carbonaceous materials-supported Pt single-atom catalysts (SACs) have attracted substantial research attention because of their maximum atom utilization efficiency and great potential to

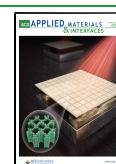
achieve high activities toward HER.^{12–21} For example, single-atom Pt on graphdiyne prepared via the thermal process exhibits mass activity up to 26.9 times higher than that of commercial Pt/C catalysts.²²

Notably, for high-efficiency HER process, it generally requires the electrode to be able to deal with fast electron and mass transportation, prompt and efficient gas release, and long-term durability.⁹ A general strategy is to construct a hybrid three dimensional (3D) electrocatalysts with hierarchical nanostructures on the surficial layer for catalysis such as dendritic structure, nanowire array, and nanosheet array.^{23,24} These nanostructures could be derived from either active species or supports, which would generate several concomitant benefits, including reduced interaction between bubbles and electrodes, increased electrochemical active surface area (ECSA), and accelerated reaction and diffusion kinetics.^{25,26}

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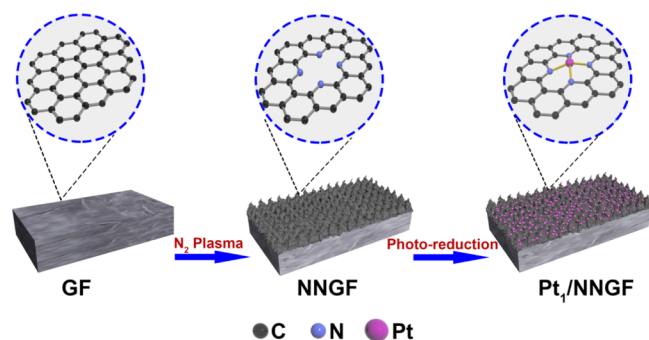
Additionally, in the case of self-supported catalysts, the absence of chemical binders enables more sufficient exposure of the active sites and thus leads to further activity enhancement.^{27–29} Unfortunately, SACs, different from conventional nanoparticles based catalysts, are limited by the atomically dispersed characteristic for preparing their own abovementioned beneficial architectures.^{30–33} Therefore, it is imperative to develop well-designed supports with large surface area, facilitated gas pathway, and high electrical conductivity for SACs to further improve their utilization for practical applications.

Here, we propose a 3D nanoarray-structured nitrogen-doped graphite foil (NNGF)-supported single-atom Pt as the free-standing HER electrode via a facile plasma treatment and photo reduction method. N₂ plasma treatment of commercial graphite foil (GF) endows NNGF the surficial layer with vertically aligned nanoarrays, abundant N dopants, and high hydrophilicity advantages. These advantages not only maximize the exposure of photoreduced single Pt atoms but also enhance the reaction and diffusion kinetics. Meanwhile, the untreated bulk layer of NNGF retains the graphitic structure, thus providing the integrated electrode with high conductivity and mechanical stability. As expected, the as-prepared Pt₁/NNGF requires a low overpotential of 0.023 V to drive a current density of 10 mA cm⁻² and exhibits a small Tafel slope of 29.1 mV dec⁻¹ as well as long-term durability.

RESULTS AND DISCUSSION

An ideal-support NNGF with both surficial nanoarray layer and bulk graphitic layer was first prepared by the one-step N₂ plasma process of the commercial GF precursor with a thickness of about 100 μm (Scheme 1). Uniform nanoarray-

Scheme 1. Schematic Illustration of Preparation of Pt₁/NNGF



structured pits (about 500 nm in diameter) are formed on the surface of NNGF (Figure 1a,b) after N₂ plasma treatment for 60 min, which is different from the smooth surface of commercial GF (Figure S1). The NNGF exhibits mesopores with a diameter of about 3.5 nm as observed by the nitrogen adsorption–desorption curves (Figure S2), which are expected to assist the adsorption and anchoring of PtCl₆²⁻ ions. Additionally, the plasma-induced nanoarrays are expected to facilitate gas bubbles releasing in gas-involved electrochemical reactions and increase ECSA for exposure of single atoms.^{25,34} As calculated, NNGF exhibits a larger electrochemical double-layer capacitance (*C*_{dl}) of 2.76 mF cm⁻² than pristine GF (0.39 mF cm⁻²) (Figure S3), suggesting the larger ECSA of NNGF. Energy-dispersive X-ray spectroscopy mapping images further indicate the existence and homogeneous dispersion of N and

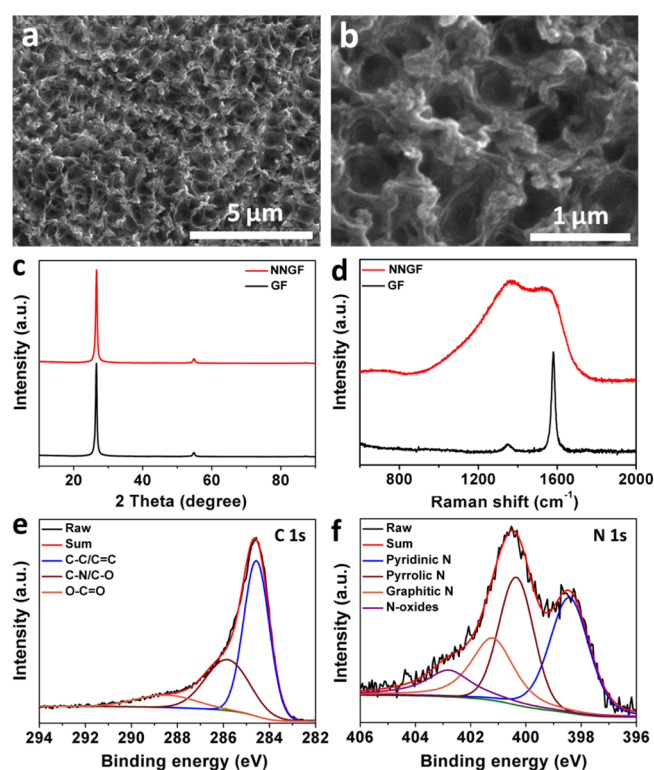


Figure 1. (a,b) SEM images of NNGF; (c) XRD patterns and (d) Raman spectra of NNGF and GF substrates; high-resolution (e) C 1s and (f) N 1s XPS spectra of NNGF.

O dopants on the surficial layer of NNGF, revealing a successful modification of GF by plasma bombardment (Figure S4). In addition, because N₂ plasma mainly focused on modifying the surface layer of GF, the untreated part of NNGF still remained integrated as the graphitic structure of GF (Figure S5a,b). Cross-sectional scanning electron microscopy (SEM) images show that the thickness of the surface layer is about 10 μm, while that of the untreated layer is 90 μm (Figure S5c,d). The mass ratio of the N-doped layer is measured to be 3.2%. The untreated graphitic structure endows NNGF the high conductivity (a low square resistance of 0.126 Ω sq⁻¹), mechanical stability, and flexibility properties (Figure S6).

The structure changes of GF after N₂ plasma treatment were first characterized by X-ray diffraction (XRD) patterns. As shown in Figure 1c, both NNGF and GF exhibit a main peak at 26.5° assigned to (002) diffraction of graphite, revealing that NNGF mostly inherits the graphitic structures from GF.³⁵ A slightly negative shift of the (002) peak of NNGF than GF is because of the formation of amorphous carbon on the surficial layer during the plasma treatment (Figure S7). To further confirm the surficial structure of NNGF, Raman spectra were also performed. As shown in Figure 1d, pristine GF showed an intensive G band (associated with graphitic structure) at 1580 cm⁻¹ and a weak D band (associated with disordered band) at 1350 cm⁻¹ with an intensity ratio of D to G peak (*I*_D/*I*_G) of 0.09, indicating a nearly entire graphitic structure.³⁶ Different from GF, NNGF exhibited much broader peaks and positively shifted D and G bands, suggesting the existence of abundant defects/disorders. The *I*_D/*I*_G ratio was determined to be 1.07, reconfirming the much greater disorder/defects in the active layer of NNGF.³⁷ These disorders on the surface are contributed to the disruption of the graphitic structure during

plasma bombardment as well as the introduction of N and O dopants. After scraping off the active layer of NNGF, it is observed that the untreated part preserved a pristine graphitic structure with an I_D/I_G of 0.09, accordingly well with our forecasts (Figure S8).

The GF precursor showed strong signals for C elements in X-ray photoelectron spectroscopy (XPS), with no peaks for N and weak signals for O dopants (Figure S9). As a comparison, the atom ratio of C, N, and O elements in NNGF was measured to be 82.4, 8.3, and 9.3%, respectively, indicating that N dopants were doped during plasma treatment (Figure S10a). The high-resolution C 1s spectrum indicates that C elements mainly exist in the form of C=O, C–O/C–N and C–C/C=C, corresponding to peaks at 288.5, 285.8, and 284.6 eV respectively, suggesting the successful modification of GF by N and O doping (Figure 1e).³⁸ The N 1s spectrum can be deconvoluted into four peaks associated with pyridinic N (398.4 eV), pyrrolic N (400.3 eV), graphitic N (401.2 eV), and negligible oxidized N (402.9 eV) (Figure 1f).^{34,39} The high contents of pyridinic N and pyrrolic N (>78%) are expected to enhance the electron-donor tendency of carbonaceous materials, thus facilitating the anchor of metallic species and improving the reaction kinetic.^{40–42} O-containing functional groups in the form of C–O (533.3 eV) and C=O (531.9 eV) groups are formed because of the exposure of NNGF to air after plasma treatment (Figure S10b).⁴³ Also, the existence of N and O dopants endows NNGF with better electrolyte wettability for facilitated mass transportation, revealed by a smaller contact angle of 5.3° than that of GF (84.6°) (Figure S11).

Single-atom Pt catalysts (denoted as Pt₁) were facilely deposited on NNGF as a self-standing electrode toward HER via a photochemical reduction of H₂PtCl₆ (Scheme 1).⁴⁰ As shown in Figure 2a,b, the nanoarray structure of NNGF is well

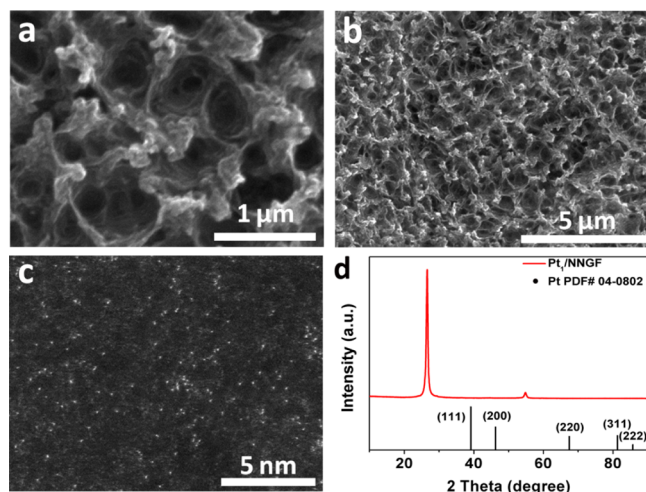


Figure 2. (a,b) SEM images, (c) HAADF–STEM image, and (d) XRD pattern of Pt₁/NNGF.

preserved after loading Pt SACs, and the pore size distribution of Pt₁/NNGF is similar to that of NNGF (Figure S12). No nanoparticles are observed in SEM or transmission electron microscopy (TEM) images (Figure S13). High-angle annular dark field scanning TEM (HAADF–STEM) image indicates that isolated Pt atoms are uniformly distributed on NNGF with a narrow size range (<0.4 nm) (Figure 2c). The Pt SACs loading is determined to be 4.96 μg cm^{−2} based on inductively

coupled plasma–optical emission spectroscopy (ICP–OES) measurement. In addition, there are no characteristic peaks of Pt crystals observed in XRD patterns (JCPDS no. 04-0802), suggesting the extremely atomical dispersion of Pt in Pt₁/NNGF (Figure 2d). Notably, Pt₁/NNGF shows a decreased I_D/I_G ratio (1.01) than NNGF in Raman spectra, suggesting that some graphitic structures are reconstructed. Such reconstruction in Pt₁/NNGF is mainly attributed to the slight reduction of O-containing groups under ultraviolet light for Pt₁ preparation (Figure S14).⁴⁴

XPS measurements were then performed to investigate the composition and chemical state of Pt₁/NNGF. The high-resolution C 1s and O 1s spectra showed decreased peaks for C–O and O–C=O, which is attributed to the partial reduction of O containing groups, in accordance with the Raman results (Figure S15).⁴⁴ Similarly, the peak for the oxidized N species in the high-resolution N 1s spectrum also decreases because of photochemical reduction. Additionally, a larger peak at 398.6 eV is observed, which is because of the formation of Pt–N bonds (Figure S15c, Table S1).^{32,37,45,46} As shown in Figure 3a, commercial Pt/C exhibits Pt 4f_{7/2} and Pt

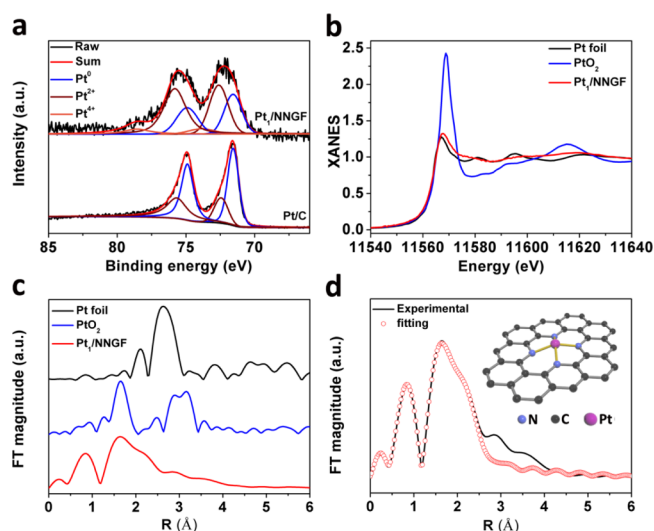


Figure 3. (a) XPS of Pt/C and Pt₁/NNGF; (b) XANES spectra at the Pt L₃-edge for Pt foil, PtO₂, and Pt₁/NNGF; (c) FT of EXAFS spectra of Pt foil, PtO₂, and Pt₁/NNGF; (d) R-space fitting curves for Pt₁/NNGF.

4f_{5/2} peaks at 71.2 and 74.5 eV, suggesting that Pt mainly exists in the form of Pt⁰. Notably, the Pt 4f peak in Pt₁/NNGF exhibits stronger peaks for Pt²⁺ and new peaks for Pt⁴⁺.^{40,47,48} To further probe the electronic structure and coordination environment of Pt₁ at the atomic level, we carried out X-ray absorption of fine structure (XAFS) measurement. The average oxidation state of Pt substances could be depicted by whiteline (WL) intensity of the sample at the Pt L₃-edge. In the X-ray absorption near-edge structure (XANES) curves, Pt₁/NNGF shows a higher WL intensity than Pt foil, also indicative of the presence of Pt^{δ+}, in well accordance with XPS results (Figure 3b). As evidenced in the Fourier transform (FT) curves of extended X-ray absorption fine-structure (EXAFS) spectra, Pt foil exhibits a main peak at 2.64 Å corresponding to the characteristic Pt–Pt bond; while Pt₁/NNGF shows no such peaks but a dominant peak at 1.67 Å ascribed to the single scattering bond of Pt–N coordination

(Figure 3c). According to the structural parameters extracted from EXAFS, the local structure of Pt is supposed to be Pt–N₄ (Figure 3d) (Table S2). Taking the morphological and structural results together, the existence of isolated Pt atoms is well confirmed.

The electrocatalytic activity of the as-obtained Pt₁/NNGF catalysts toward HER was studied in N₂-saturated 0.5 M H₂SO₄ solution with a standard three-electrode system via the linear sweep voltammetry (LSV) method without *iR* compensation. As shown in Figure 4a, the Pt₁/NNGF

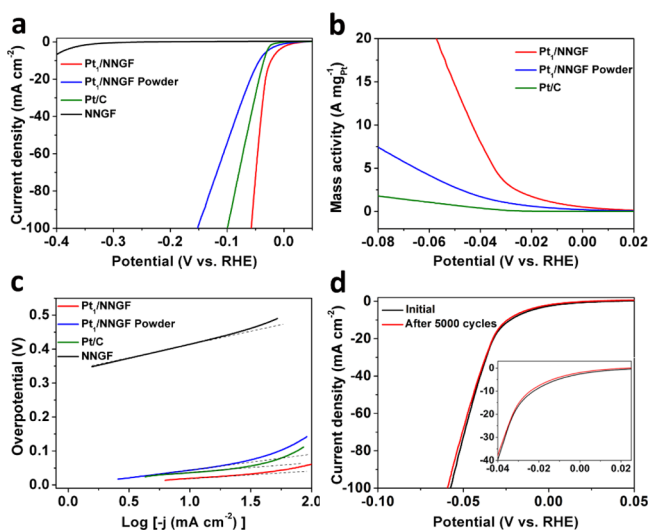


Figure 4. (a) HER polarization curves, (b) mass activities, and (c) Tafel plots of Pt₁/NNGF, Pt₁/NNGF powder, and Pt/C and NNGF catalysts; (d) polarization curves of Pt₁/NNGF before and after ADT for 5000 CV cycles; inset shows the enlarged curves.

electrode exhibits superior activity with a low overpotential of 0.023 V at a current density of 10 mA cm⁻², which is much better than pure NNGF, commercial Pt/C catalysts (0.035 V at 10 mA cm⁻²), and comparable with other reported single-atom Pt-based catalysts (Table S3).^{22,30,40,41,49–54} Remarkably enough, the Pt₁/NNGF powders scraped off from the Pt₁/NNGF electrode exhibit a weaker HER activity with an overpotential of 0.043 V to drive 10 mA cm⁻². The higher catalytic activity of the Pt₁/NNGF electrode than Pt₁/NNGF powders is ascribed to the surficial nanoarrays which not only allow adequate exposure of active sites to electrolyte but also facilitate gas releasing to protect Pt species from detachment. As shown in Figure S16, Pt₁/NNGF powders show a much attenuated *C_{dl}* of 1.22 mF cm⁻² than the Pt₁/NNGF electrode (3.11 mF cm⁻²), suggesting a dramatically decreased ECSA. Also benefited by these surficial nanoarrays, Pt₁/NNGF electrode exhibits a low overpotential of 0.058 V at a higher current density of 100 mA cm⁻². Figure 4b shows the mass activities of these Pt-based catalysts. The Pt₁/NNGF electrode exhibits a high mass activity of 14.64 A mg_{Pt}⁻¹ at an overpotential of 0.05 V, which is 3.86 and 16.6 times larger than Pt₁/NNGF powder and Pt/C catalysts. The H₂ turnover frequency of the Pt₁/NNGF electrode was calculated to be 14.1 s⁻¹ at an overpotential of 0.05 V. Therefore, the enhanced performance of the Pt₁/NNGF electrode is also attributed to the formation of the isolated Pt atoms. The Tafel plots of corresponding electrodes were also recorded (Figure 4c). The Pt₁/NNGF electrode exhibits a Tafel slope of 29.1 mV dec⁻¹, smaller than Pt/C (29.5 mV dec⁻¹), NNGF (83.0 mV dec⁻¹),

and Pt₁/NNGF powder (44.4 mV dec⁻¹), indicating a faster kinetic.

In addition, the long-term stability of the Pt₁/NNGF electrode was evaluated via the accelerated degradation test (ADT) for 5000 cycles. As shown in Figure 4d and S17, Pt₁/NNGF exhibits a negligible degradation of current density (2.2%) at an overpotential at 0.05 V after the ADT, while commercial Pt/C catalysts and Pt₁/NNGF powder suffer from severe current density losses of 28.9 and 9.2% respectively. After 5000 cyclic voltammetry (CV) cycles, there is no visible changes for the Pt₁/NNGF electrode (Figure S18a), and the surficial nanoarray structures are well preserved (Figure S18b,c), revealing the mechanical stability of the synthesized NNGF support. Besides, the HAADF–STEM image indicates that Pt atoms are still atomically dispersed on NNGF, which is mainly attributed to the stable Pt–N₄ structure (Figure S18d). The chronopotentiometric measurements were also recorded for 100,000 s at an overpotential at 0.05 V, suggesting an enhanced stability of the Pt₁/NNGF catalyst in comparison with Pt/C catalysts (Figure S19). The outstanding activity and durability of Pt₁/NNGF is attributed to four following reasons: (1) the surficial layer of NNGF not only improves abundant exposure of Pt SACs but also enhances the reaction and diffusion kinetics to reduce the negative effect of gas bubbles on the catalyst; (2) the untreated layer in NNGF provides the required high electrical conductivity and mechanical stability; (3) the constructed Pt–N₄ structure endows the catalyst high intrinsic activity and stability because the prevented agglomeration of active Pt sites; and (4) the binder-free characteristics further improve the utilization of Pt SACs because of the accessible active sites to electrolyte.

CONCLUSIONS

In summary, we have rationally designed an NNGF to support single-atom Pt catalysts for direct application in HER catalysis. This as-prepared Pt₁/NNGF electrode exhibits high HER activity with a low overpotential of 0.023 V at a current density of 10 mA cm⁻² and a small Tafel slope of 29.1 mV dec⁻¹ as well as excellent long-term durability. The outstanding performance of the Pt₁/NNGF electrode is mainly ascribed to the surficial nanoarrays of NNGF for increased ECSAs and facilitated reaction and diffusion kinetics, the retained graphite structure of NNGF for high electrical conductivity and mechanical stability, the constructed Pt–N₄ structure for high intrinsic activity and stability, and the binder-free characteristic for further enhanced utilization of SACs.

EXPERIMENTAL SECTION

Preparation of NNGF. NNGF was prepared from the commercial GF precursor (Beijing Jingtian Inc.) via N₂ plasma treatment using a LED-2000 system (Chengdu Tongchuang Inc.).³⁴ In detail, a large piece of GF (e.g. 5 × 5 cm²) was first washed by ethanol, acetone, and deionized water and then dried in a vacuum oven. Then, GF was put into the chamber to be treated with N₂ plasma for 30 min. The applied voltage was set as 60 kV, and the pressure of N₂ was 0.01 Pa. Afterward, the as-obtained NNGF was cut into pieces of 1 × 1 cm² for further use.

Preparation of Pt₁/NNGF. Isolated Pt atoms were deposited on NNGF by a photochemical reduction process.⁴⁰ In detail, 30 μL of 1 mM H₂PtCl₆ alcohol solution was dropped on NNGF (1 × 1 cm²) and then dried in air. The as-obtained pieces were then irradiated under a mercury lamp with a distance of 10 cm for 1 h, and the adsorbed PtCl₆²⁻ ions were reduced to Pt atoms. The as-obtained pieces were washed by deionized water to remove the residual ions. The

obtained Pt₁/NNGF pieces were directly used as the working electrode without any further treatment.

Characterization. The morphology of NNGF and Pt₁/NNGF electrode was characterized with a field emission Sirion-200 SEM at 20 kV. TEM images were obtained using a Tecnai F20 TEM instrument. The HAADF-STEM images were taken by a G280-300 ChemiSTEM microscope at 300 kV. The mass of NNGF and the N doped layer was measured by MS205DU (Mettler Toledo N315) using a large piece of NNGF (25 cm²). Brunauer–Emmett–Teller tests were conducted by ASAP 2460 (Micromeritics Inc.) with N₂. XRD patterns were obtained using a Y-2000 X-ray diffractometer (Dandong Aolong Instruments) with Cu K α radiation ($\lambda = 0.15418$ nm). Raman spectra were measured on an Omega XR Raman microscope with a 532 nm laser. XPS was carried out on a PHI 5000 XPS system with an Al K α X-ray source (1486.6 eV photons). Elemental analysis was conducted by ICP–OES. The XAFS measurements were collected on the beamline BL01C1 in NSRRC and were provided technical support by “Ceshigo Research Service, Agency for XAS and DFT/MD simulation (www.ceshigo.com)”. The radiation was monochromatized by a Si(111) double-crystal monochromator. XANES and extended EXAFS data reduction and analysis were processed by Athena software.

Electrochemical Tests. Electrochemical measurements were performed on a CHI 760D electrochemical workstation in an H-type electrolytic cell with a Nafion membrane. A three-electrode system was used with the Pt₁/NNGF electrode, platinum mesh, and saturated silver/silver chloride electrode (Ag/AgCl) electrode as working, counter, and reference electrodes, respectively. The C_{dl} was measured in 0.5 M H₂SO₄ in a non-Faradaic region (0.06–0.16 V vs RHE) by a CV method with scan rates of 1, 2, 5, 10, 20, 40, 60, and 80 mV s^{−1}. LSV curves were recorded in N₂-saturated 0.5 M H₂SO₄ solution at a scan rate of 1 mV s^{−1} without iR-compensation. The ADT were performed in a potential range of 0.4 to −0.15 V (vs RHE) for 5000 CV cycles at a scan rate of 100 mV^{−1} in 0.5 M H₂SO₄. Chronoamperometry measurements were conducted at −0.05 V for 100,000 s.

As the counterpart, Pt₁/NNGF powders were prepared as follows: the surficial substances of the Pt₁/NNGF electrode (1 × 1 cm²) were scraped off by a blade and dispersed in a solution of 80 μ L of water and 20 μ L of alcohol. The as-prepared dispersion was mixed with 5 μ L 5% Nafion (Alfa Aesar) and sonicated for 30 min. The resulted dispersion was then all drop-cast on the GF substrate (1 × 1 cm²) to obtain the Pt₁/NNGF powder electrode.

In comparison with the Pt₁/NNGF electrode, 20% Pt/C catalysts (Alfa Aesar) were drop-cast on NNGF with a loading of 40 μ g_{Pt} cm^{−2}. In detail, 5 mg of Pt/C catalysts was dispersed into a solution of 760 μ L of water, 190 μ L of ethanol, and 50 μ L of 5 wt % Nafion by sonication. Then, 40 μ L of catalyst ink was drop-cast on the NNGF substrate. The as-prepared pieces were dried at room temperature.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.0c09615>.

SEM, CV, and XPS of GF and NNGF; XPS of Pt₁/NNGF; and stability tests of Pt₁/NNGF, Pt₁/NNGF powder, and Pt/C (PDF)

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Author Contributions

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Notes

The authors declare no competing financial interest.

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